

HVAC parts list - 4407 Aldrich Rd, Bellingham WA 98226

SFR2025-00084 model **4407-aldrich-hvac** printed 2026-09-25 ducted heat 33,420 Btuh cooling 1.06 t 1 outdoor 0 heads 1 ducted units 21 registers line sets 34 ft duct 449 ft vent 28 ft 12 days on site
Design 19 F / 78 F (bellingham: WSEC Table RC-1 Bellingham IAP 19 F / 78 F [S]; MJ8 Table 1A 23 F / 76 F [V]). Foundation crawl, line sets concealed, ducts crawl_vented, ventilation exhaust. Quantities only - the prices are set on the price list in the app.

Loads

heating 33,420 Btuh (12.3 Btuh/sf) · cooling 11,504 + 1,159 latent · WSU sheet 35,536 Btuh · infiltration 5 ACH50 -> 0.448 ACH at design · ventilation 85.9 cfm (exhaust)

room	storey	sf	heat Btuh	cool Btuh	glass sf	ext wall ft
MASTER BEDROOM	upper	254	4,943	1,802	70	64
LAUNDRY	main	126	4,741	1,061	66	64
DEN ROOM	main	417	4,355	411	36	52
BEDROOM 2	main	191	2,794	1,285	69	28
LIVING ROOM	upper	136	2,384	1,380	43	28
DINING ROOM	upper	164	2,064	761	30	25
ENTRY	main	112	1,879	1,249	46	20
KITCHEN	upper	165	1,819	2,289	19	23
BEDROOM 1	main	184	1,608	252	0	24
MASTER BATH	upper	115	1,310	133	0	20
STAIRS	upper	286	1,139	564	10	10
MASTER W.I.C.	upper	60	366	64	0	5
BATH	upper	25	349	30	0	6
BATH	main	40	324	4	0	5
BATH	main	40	316	4	0	5
W.I.C.	main	18	230	3	0	4
PANTRY	upper	20	213	23	0	3
HALL	main	63	84	0	0	0
OFFICE	main	53	72	0	0	0
HALLWAY	main	37	56	5	0	0

The system

unit house - ducted: heat 33,420 Btuh, capacity at design 32,906 Btuh

outdoor eq_hp_42 42k (central) on the W wall, at the electrician's Heat pump condenser (DEN ROOM, electrical/jobs/aldrich/devices.json); 35 ft of line set
indoor eq_ahu_42 42k in LAUNDRY (main), 1028 cfm

register room	storey	cfm	duct
BATH	main	30	5 in
DEN ROOM	main	66	6 in
DEN ROOM	main	66	6 in
ENTRY	main	52	6 in
ENTRY	main	52	6 in
HALLWAY	main	30	5 in
HALL	main	30	5 in
BEDROOM 1	main	49	5 in
BEDROOM 2	main	53	6 in
BEDROOM 2	main	53	6 in
MASTER BEDROOM	upper	75	6 in
MASTER BEDROOM	upper	75	6 in
LIVING ROOM	upper	57	6 in
LIVING ROOM	upper	57	6 in
DINING ROOM	upper	63	6 in
KITCHEN	upper	63	6 in
KITCHEN	upper	63	6 in
KITCHEN	upper	63	6 in
STAIRS	upper	47	5 in
BATH	upper	30	5 in
MASTER BATH	upper	40	5 in

return	storey	cfm	duct	grille
ENTRY	main	481	12 in	14x20
DINING ROOM	upper	634	14 in	16x20

Runs (measured, before waste)

system / size	LF
dryer 4	5
exhaust 4	12
exhaust 6	8
hood 6	4
refrigerant 0.375x0.75	34
return 12	20
return 14	20
supply 14	10
supply 5	130
supply 6	180
trunk 12	44
trunk 14	45

Equipment

Equipment	measured	order	unit	note
item				
Air handler - variable speed, 42k (3.5 ton)	1	1	EA	air handler 42k in LAUNDRY (main; job.json (main/LAUNDRY))
Backup heat strip 5 kW (controlled per WSEC R403.1.2)	1	1	EA	backup / defrost tempering 5 kW (heat-pump shortfall at design 514 Btuh; Manual S: <= 5 kW while it is under 1
Heat pump outdoor unit - centrally ducted 42k (3.5 ton), cold-climate variable capacity	1	1	EA	ducted heat pump outdoor unit 42k (~32,906 Btuh at 19 F), at the electrician's Heat pump condenser (DEN ROOM,
Thermostat - programmable, heat pump with aux-heat lockout (WSEC R403.1.1 / R403.1.2)	1	1	EA	WSEC R403.1.1 programmable; heat-pump aux lockout R403.1.2

Rough-in

Condensate	measured	order	unit	note
item				
Condensate line 3/4 in PVC (or 5/8 vinyl with the line set)	8	9	LF	air handler condensate to an approved discharge
Condensate trap + cleanout tee	1	1	EA	trap + cleanout tee

Ductwork	measured	order	unit	note
item				
Duct hanger strap / saddle (flex every 4 ft, rigid every 10 ft)	100	100	EA	trunk
Duct insulation R-8 wrap (rigid duct outside the envelope, WSEC R403.3.1)	43.9	49	LF	main trunk in the vented crawl - R-8 (WSEC R403.3.1)
Duct mastic (gal) + UL-181 foil tape	5	5	EA	mastic by the gallon per 100 ft of duct
Flex duct R-8 5 in	130.3	137	LF	BATH: 30 cfm · HALLWAY: 30 cfm · HALL: 30 cfm · BEDROOM 1: 49 cfm · STAIRS: 47 cfm · MASTER BATH: 40 cfm
Flex duct R-8 6 in	179.6	189	LF	DEN ROOM: 66 cfm · ENTRY: 52 cfm · BEDROOM 2: 53 cfm · MASTER BEDROOM: 75 cfm · LIVING ROOM: 57 cfm · DINING R
Register boot (4x10 / 4x12 floor or 6x10 ceiling)	21	21	EA	
Return duct 12-16 in (panned joist not allowed for a new return)	40.2	43	LF	main central return 12 in, 481 cfm, 14x20 filter grille · upper central return 14 in, 634 cfm, 16x20 filter gr
Supply + return plenums (fabricated, sealed, lined where required)	1	1	EA	supply + return plenums at the unit
Supply trunk - galvanized rectangular / 10-14 in round (extended plenum)	101.4	107	LF	main: 12 in equivalent, 481 cfm - reducing past 24 ft · upper: 14 in equivalent, 634 cfm - reducing past 24 ft
Takeoff / starting collar with balancing damper	21	21	EA	
Transfer grille pair / jumper duct (closed-door pressure relief)	4	4	EA	DEN ROOM: 132 cfm needs ~72 sq in free area at 3 Pa - transfer grille pair or jumper duct · BEDROOM 1: 49 cfm

Refrigerant piping

item	measured	order	unit	note
Anti-vibration pads (set of 4)	1	1	EA	
Communication cable 14/4 stranded (outdoor unit to each indoor unit)	34.5	38	LF	with the line set
Equipment pad - composite, sized to the outdoor unit	1	1	EA	outdoor unit pad
Line set 3/8 x 3/4 insulated	34.5	38	LF	42k indoor unit, 35 ft
Outdoor unit stand / wall bracket (snow and drainage clearance)	1	1	EA	snow / drainage stand
Wall penetration - 3 in sleeve, sealed and flashed	1	1	EA	rim penetration for the line set

Support + protection

item	measured	order	unit	note
Fire caulk / sealant (tube)	1	1	EA	fire caulk / sealant at the penetrations

Ventilation and exhaust

Ventilation

item	measured	order	unit	note
Bath fan exhaust duct 4 in insulated flex (50 cfm fan: <= 25 ft, Table M1505.4.4.2)	11.5	13	LF	BATH: 50 cfm, 6 ft, 2 elbows -> 4 in flex (Table M1505.4.4.2 allows 25 ft)
Dryer duct 4 in 90 elbow	2	2	EA	
Dryer duct 4 in rigid metal, smooth wall (M1502.4.2)	5	6	LF	from the electrician's dryer outlet in LAUNDRY: 5 ft
Dryer vent box (recessed)	1	1	EA	
Dryer wall cap with damper, no screen (M1502.3)	1	1	EA	
Exhaust duct 6 in insulated (80-110 cfm fans, whole-house fan: Table M1505.4.4.2)	7.7	9	LF	BATH whole-house fan: 96 cfm, 6 ft, 2 elbows -> 6 in flex (Table M1505.4.4.2 allows 45 ft) · MASTER BATH: 110
Outdoor air inlet 4 sq in NFA, through-wall with screen (WA exhaust-only whole-house ventilation, one per habitable room)	5	5	EA	unit house: an outdoor-air inlet (4 sq in NFA) in each of 5 habitable spaces (WA exhaust-only systems) - window
Range hood duct 6-10 in rigid, sealed (M1503.3)	4	5	LF	6 in rigid for a 300 cfm hood at the electrician's range outlet (electrical/jobs/aldrich/devices.json) (M1503.
Range hood wall / roof cap with damper	1	1	EA	6 in wall cap with damper
Wall / roof cap 6 in with backdraft damper	2	2	EA	BATH whole-house fan: wall cap with backdraft damper · MASTER BATH: wall cap with backdraft damper
Wall cap 4 in with backdraft damper	2	2	EA	BATH: wall cap with backdraft damper
Whole-house exhaust fan, continuous duty, <= 1.0 sone (M1505.4) - replaces one electrical allowance fan	1	1	EA	unit house: whole-house fan 86 cfm continuous (Qr 57 x Csystem 1.5, one fan = unbalanced, not distributed) in
Whole-house ventilation control - 24 h timer / override, labelled (WA M1505.4.2)	1	1	EA	labelled "WHOLE HOUSE VENTILATION - LEAVE ON" 24 h control with manual override (WA M1505.4.2); the electrician

Trim

Registers + grilles

item	measured	order	unit	note
Filters MERV 8-13 (first set)	1	1	EA	first filter set
Floor register 4x10 / 4x12	21	21	EA	BATH · DEN ROOM · ENTRY · HALLWAY · HALL · BEDROOM 1
Return / filter grille 20x25 (or 14x30)	2	2	EA	ENTRY (main): 14x20 at <= 300 fpm · DINING ROOM (upper): 16x20 at <= 300 fpm

Start-up, tests and design

Commissioning

item	measured	order	unit	note
Air balance of supply registers (per ducted system)	1	1	EA	register balance
Duct leakage test, total cfm25 (WSEC R403.3.5) - certified tester	1	1	EA	ducts outside the thermal envelope (vented crawl): total leakage test (WSEC R403.3.5)
Equipment labels, WSEC certificate data, owner hand-over	1	1	EA	per dwelling unit
Load calculation + equipment selection for the permit (Manual J / S or the WSU sizing form)	1	1	EA	room-by-room loads for the permit (this takeoff is the draft)
Start-up, charge verification, controls set-up (per outdoor unit)	1	1	EA	42k system
Ventilation flow test - whole-house + local exhaust (WA M1505.4.3 / M1505.4.4)	1	1	EA	unit house: whole-house 86 cfm + every local exhaust measured at the grille (WA M1505.4.3 / M1505.4.4)

Refrigerant piping

item	measured	order	unit	note
Nitrogen pressure test + evacuation to 500 micron (per system)	1	1	EA	per refrigerant system
Refrigerant - additional charge beyond the pre-charge (lb)	0.4	0.4	lb	5 oz past the pre-charge [U: the maker's oz / ft]

The checker

PRACTICE electrical: the 42k central outdoor unit typically needs a 35 A circuit [U: the nameplate MOCOP governs]; the electrical job carries 30 A
note: tracer fragments merged into one space (same name, touching): BATH (main: 2 pieces, 80 sf)
note: under 70 sf (IRC R304.1 - not a habitable room), so carried by a neighbour and never given a head or an ERV supply: OFFICE 53 sf
note: the 42k carries 0.98 x the heating load at 19 F - inside the 5 % the takeoff accepts; the strip carries the rest
note: bath fans: 4 read from the electrical skill (electrical/jobs/aldrich/devices.json) - the electrician supplies and wires them; HVAC ducts and caps each one
note: unit house: the plan's 96 cfm whole-house fan covers the 86 cfm WA rate (Qr 57 x 1.5 for one exhaust fan)
note: equipment must meet HSPF2 >= 9.4 (the WSEC credit); design temperature 19 F <= 23 F: a NEEP-listed cold-climate variable-capacity unit (the 3.6 note)
note: ducts in the vented crawl are outside the thermal envelope: R-8 on every duct, mastic-sealed, and the total duct leakage test before final
note: refrigerant: a split system installed from 1/1/2026 is an A2L (R-454B / R-32) unit under the EPA Technology Transitions rule; R-410A equipment is out. A2L units carry leak detection / mitigation per their listing - it rides with the equipment line
note: electrical: one of its bath-fan allowances becomes the HVAC whole-house fan (supplied here) - strike it from the electrical allowance, keep the wiring and add the 24 h control

Labour and inspections

crew of 2: set 10.4 h / 2 d, rough 50.6 h / 8 d, trim 3.3 h / 1 d, startup 6.5 h / 1 d - 12 days at 6.25 h. crew minutes are PLACEHOLDERS
(no task-level install hours were found: references/research-hvac-market.md 3.3) - set them from a sub or the crew
Mechanical rough-in (cover) - after framing, before insulation; line sets pressure-tested and in place; ducts, boots, exhaust ducts and caps; firestops at the garage separation
Duct leakage test - rough-in or final; total leakage at 25 Pa per WSEC R403.3.5 - waived only when every duct and the air handler are inside the envelope
Ventilation flow test - before final; whole-house rate and every local exhaust measured (WA M1505.4)
Final mechanical - equipment set, started, labelled; equipment listing / HSPF2 on the certificate, controls, condensate, clearances
permit: Whatcom County PDS residential mechanical fee schedule - NOT FOUND in the 9/25 research (references/research-hvac-code.md 8): price it from the county sheet